How disrupted spatial fluxes reshape food web structure along the river–sea continuum

Ismaël Lajaaiti, Eric Edeline, Anik Brind'Amour, Alain Danet

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Abstract

Anthropic disturbances and ecosystem processes transcend the artificial boundaries between land, freshwater, and sea. While these systems are interconnected through species and material fluxes, they are often studied in isolation. In response, the concept of metaecosystem has emerged as a powerful framework to understand how fluxes among ecosystems shape their functions and dynamics. Disrupting these fluxes can have dramatic consequences that ripple across entire landscapes. In aquatic systems, dams and nutrient pollution are known to strongly affect fish communities. Yet, little is still known about how these effects propagate and potentially accumulate, through hydrographic basins and ultimately to the sea. Using bayesian spatio-temporal statistical models (R-INLA), we aim to analyse existing data on food-web structure across France, combined with environmental variables (dams, nutrients, …), to quantify how breaking or increasing fluxes among systems disturbs food-web structure locally and how these disturbances spread throughout the metaecosystem.

Keywords: *foodweb ecology, stream network, spatial autocorrelation, bayesian statistics.*

Introduction

While patterns in community change within river network have been well studied, these studies have mostly focus on specise richness (Bonnaffé et al. 2024a), or roughly defined functional composition (Vannote et al. 1980).

- Barnes et al. (2026) food web complexity mediates diversity-functioning relationship. Interesting metric of trophic complementarity, could be used for our study if we can formulate clear hypothesis of stressors on that metric.

River-estuary-sea continuum

- Xenopoulos et al. (2017) highlights the importance of accounting for the full gradient from headwater to the sea. Freshwater and coastal ecosystems are connected by flux of carbone and nutrients (N, P), as well as, movement of migratory species. Reviews work connecting river and sea, that said, most of them focus on chemical fluxes, remain relatively local, and none of them focused on the food web structure.
- Catchment land-use impacts macro-invertebrates along the river-sea continuum with no dilution effect (Bierschenk et al. 2017).
- The Land-to-Ocean Continuum (LOAC) concept has been used to study carbon fluxes between land and ocean, and how these fluxes, and feedback loop influence the carbon storage by oceans (Regnier et al. 2022).
- Hong et al. (2015) Editorial overview of studies investigating the impact of human pressure along the river-estuary-sea continuum, more focused

toward biochemistry than ecology.

- Ouellet et al. (2022) provides a conceptual model (DWOC) to advance a more holistic management of diadromous species across the watersheds-ocean continuum.
- Tee et al. (2021) studies the river-to-sea continuum of microbial communities through community composition and their function (nutrient cycling).
- Whitfield et al. (2024) reviews how species richness changes along the salinity gradient in estuaries. As we go to freshwater to brackish water we loose species richness, and as we to brackish water to the sea we recover species richness.
- Elliott and Quintino (2007) argues that it is difficult to detect the effect of anthropogenic pressure in estuary as estuaries are already stressed environment. In particular, this means thinking about the indicators used to describe ecosystem states (diversity, and functionning).
- Little et al. (2017) Lechêne et al. (2018) Olli et al. (2019) and Freitas and Araújo (2025) study how communities change along environmental gradient in estuaries.
- Attrill (2002) critiques Remane diagram and proposes and quantitive, testable alternative that consider salinity ranges instead of salinity.
- Macedo et al. (2021) Food webs are simpler during the dry season than the rainy season in the estuarine and neritic zones. During the rainy season, the food web of the neritic is largely shaped by migratory species from the estuarine and oceanic areas.
- Álvarez-Romero et al. (2015) proposes an integrated framework to include land-use change in cactchement in the impact on coastal biodiversity.
- Truchy et al. (2026) links metaecosystem and ecosystem services theories to develop an unified framework for managing sociohydroecosystems.

Freshwater ecosystems

- The River Continuum Concept has shaped the freshwater ecology researsh by links physical gradients along rivers to predictable changes in community structure, energy sources, and ecosystem functioning (Vannote et al. 1980; Doretto et al. 2020).
- Serial Discontinuity Concept is a framework to predict the impact of discontinuities, such as dams, on biotic and abiotic variables along the river continuum (Ward and Stanford 1995).
- The Flood Pulse Concept (FPC) complements the RCC as it focuses on the lateral dimension of rivers and its role in explaining community

productivity, while the RCC focused on the longitudinal aspect. The FPC states that its the rythm of flood and recession – that is, the pulse – which structures ecosystem productivity, not the flow (Junk et al. 1989).

  - Terrestrial organic matter (leaf litter, vegetation, soil nutrients) gets dissolved and mobilized into the water
  - Aquatic plants and algae explode in productivity in the shallow, sunlit floodwaters
  - Fish and invertebrates move onto the floodplain to feed and reproduce
  - When waters recede, nutrients and organic matter flush back into the main channel

- The River Wave concept as unifying framework of different concepts of freshwater ecology (Humphries et al. 2014).
- Community richness and hence food web structure is expected to be driven by nutrient availability (Ho et al. 2023).
- Carraro and Ho (2025) link river shapes community diversity and food web structure using a numerical approach. Higher spatial heterogeneity and lower biomass in elongated rivers than compact ones, driven by higher variability in nutrient input load.
- Harris et al. (2026) mobile consumers (fish, birds) use heterogeneous resources in braided river, New Zealand.

Multiple stressors

- Synergistic and antagonistic effects of multiple stressors (Jackson et al. 2021; Orr et al. 2024).
- Temperature can interact with nutrients and impact food web structure in a nonlinear way Binzer et al. (2012).
- (Danet et al. 2021; Bonnaffé et al. 2024a).
- Zhang et al. (2026) precipitation modulates human impacts on riverine food webs

Connectivity

- Small dispersal stabilizes through spatial structure and asynchrony (LeCraw et al. 2014; McCann et al. 2005; Pillai et al. 2011).
- A metaecosystem model shows that dispersal of living organisms (that consume nutrients) is stabilising,while dispersal of non-living compartments such as nutrients or detritus is destabilising (Gounand et al. 2014).
- Prey switching of mobile top predators can stabilize connected food webs through 'spatial coupling' (LeCraw et al. 2014; McCann et al. 2005; Pillai et al. 2011).

- At high dispersal/connectivity the stabilizing effect wanes as strongly connected patches behave as a single patch (LeCraw et al. 2014; McCann et al. 2005).
- Studies between pairs of the following connectivity, complexity and stability are common. However, studies investigating how these are interconnected are rare (LeCraw et al. 2014).
- Connectivity has 4 dimensions: longitudinal, lateral, vertical and temporal (Grill et al. 2019).
- We identified five pressure factors to represent the main human interferences within the four dimensions of fluvial or river connectivity (Grill et al. 2019):
  (1) river fragmentation (longitudinal);
  (2) flow regulation (lateral and temporal);
  (3) sediment trapping (longitudinal, lateral and vertical);
  (4) water consumption (lateral, vertical and temporal);
  (5) infrastructure development in riparian areas and floodplains (lateral and longitudinal).
- In freshwater system, fish dispersal is limited. A meta-analysis found that the mean dispersal kernel is around 10km (Comte and Olden 2018).
- Neutral metacommunity model explains biodiversity pattern in the Mississipi river, showing the key role of dispersal in explaining community structure along the river continuum (Muneepeerakul et al. 2008).
- Connectivity structures and mititgates effect of drying events in macroinvertebrates communities (Chalmandrier et al. 2025).
- It is key to adopt a network perspective to understand how diversity is created and maintained within riverine ecosystems (Altermatt 2013). With this perspective one can account for the dendritic network and so the spatial processes that shape diversity.
- From points (Death and Winterbourn 1995)… to lines (RCC, SDC)… to dendritic networks (Shepherd and Ellis 1997).
- River network characteristics, local network properties and their interaction were important individual contributors to explaining local species richness of aquatic insects (Altermatt et al. 2013). Effect of connectivity on richness supported experimentally (Carrara et al. 2012).
- Altermatt and Fronhofer (2018) in exprimental microcosms population densities are found to be more heterogeneous in dendritic than linear networks.

Decomposing spatio-temporal patterns

- Frelat et al. (2022) decompose change in food web structure and species composition across space and time using a tensor decomposition. We could probably try the method for our work.
- Kortsch et al. (2021) investigates change in food web in the Baltic Sea using weighted and unweighted structural metrics.

Dams

- Decrease river connectivity / reduce migration (Dean et al. 2023; He et al. 2024; Pedersen et al. 2012).
- Reservoir, lotic from lentic conditions, can result in hypoxic conditions. Favor phytoplancktons and macrophytes (Ding et al. 2026; Dean et al. 2023). Facilitate the spread of invasive species (Johnson et al. 2008).
- Reservoir increase residence time of nutrients upstream and decrease the downstream output, which can result in a decrease in primary productivity.
- Hydropeaking - that is, the sudden release of large water to answer the higher electric demand - can harm riverine community due to sudden change in water level and water flow.
- Cumulative impact of dams can reduce the nutrient input in deltas (Dean et al. 2023).
- A review (Ellis and Jones 2013) suggests that two recovery gradients exist in regulated river:
    1) a longer, thermal gradient of around 100 km;
    2) a shorter resource subsidy gradient of around 1-4 km.
- Zhu et al. (2024) shows that dam areas show lowest richness and highest connectance, while areas with dam removed show highest richness but lowest connectance.
- Dolan et al. (2026) Meta-analysis shows that native and non-native species increased after barrier removal, but the positive effect was strongest for native species
- Chan et al. (2025) dam have negative impact on diadromous fish species. The installation of fish passes didn't have a positive effect. Dam removal had a positive effect.

Nutrients

- Mwanake et al. (2026) DOC-to-nutrient ratios in river decline worldwide
- Ma et al. (2026) finds that human impact destabilize riverine fish communities, but habitat complexity buffers this destabilizing effect.

Meta-ecosystems

- Loreau et al. (2003) fundational paper that defines the concept of meta-ecosystem.
- Polis et al. (1997) founding synthesis of how spatial subsidy, flux of consumer and resources reorganise food webs.
- Nakano and Murakami (2001) shows the seasonal coupling between a forest and stream, where subidies flux alternate between season. Flux from aquatic to terrestrial peaked in spring, while the reciprocal flux peaked in summer.
- Leibold et al. (2004) provides a definition for the concept of meta-community and present different paradigms: patch dynamic, species sorting, mass effect and neutral perspective.
- Spaak et al. (2026) provide a metric of coupling between ecosystems based on the Jacobian.
- Peller et al. (2026) Nested flows of nutrients and consumers stabilize metaecosystems through cross-scale source-sink dynamics.

Traits

> **Note**
>
> This section may not be relevant for the current manuscript. It may however be useful resource for future extension of the project that may include trait space analysis.

- Brosse et al. (2021) FISHMORPH database which contains 10 morphological traits for around 50% of freshwater fish species worldwide.
- Su et al. (2023) in North American freshwater, invasive species are often highly fecund, long-lived and large. Invaded community on the other hand exhibit a high functional distance between the species it comprises leaving empty functional space for potential invaders.
- Toussaint et al. (2018) Non-native species led to marked shifts in functional diversity of the world freshwater fish faunas

Hypotheses

Building on the causal structure below (Fig. 1) and the literature reviewed above, we formulate four hypotheses.

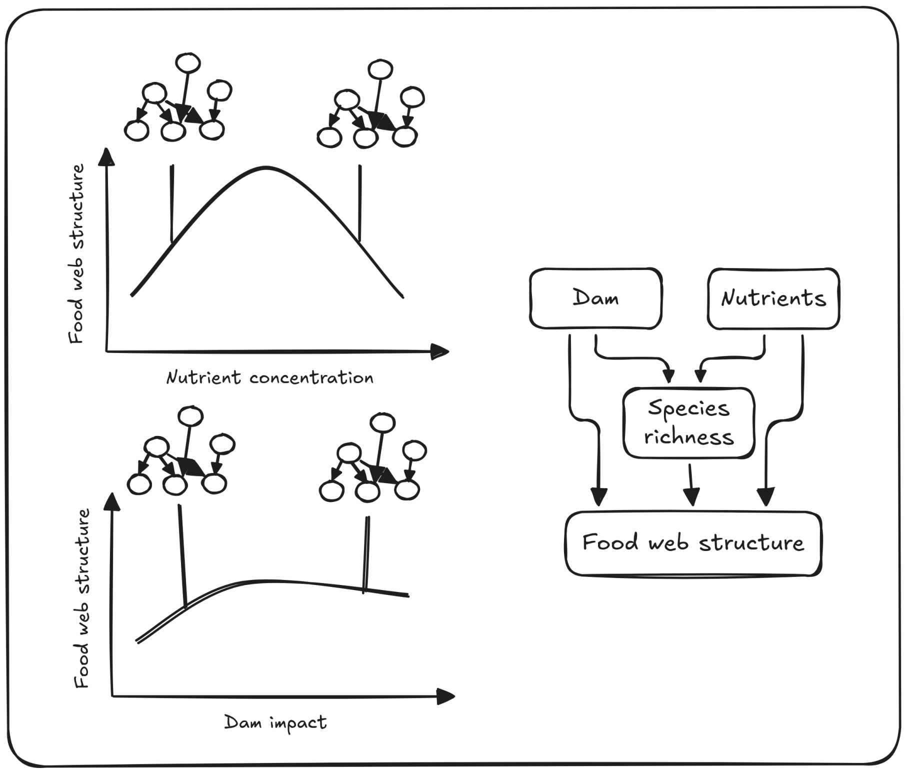

Figure 1. Causal DAGs illustrating our hypothesis of dam and nutrients impacts on food web structure, potentially mediated by species richness.

We focus on three structural metrics with well-developed theoretical expectations: **connectance**, **mean trophic level**, and a **trophic complementarity** metric following Barnes et al. (2026). Because connectance, and potentially other structural metric, are mechanically linked to species richness (Cohen and Briand 1984; Martinez 1992) we will incorporate richness as mediator variable to separate direct and richness-mediated effects .

H1: Nutrients have a unimodal effect on food web structure

We expect an overall humped relationship between nutrient availability (proxy of productivity) and food web structure (Mittelbach et al. 2001). Moderate nutrient enrichment should increase local productivity and prey availability, supporting more diverse and complex food webs (Ho et al. 2023; but see Post 2002). Beyond a threshold, excess nutrients are expected to degrade food web structure through hypoxia, algal dominance, and turbidity (Ding et al. 2026; Dean et al. 2023), or through the destabilization of higher trophic level as predicted by the *paradox of enrichment* (Rosenzweig 1971). This is consistent with Bonnaffé et al. (2024b)'s recent finding that temperature and nutrient interact magnifying food web simplification. In sum, we expect mean trophic level and trophic complementarity to rise then fall with nutrient concentration, and connectance to follow the opposite pattern (mostly through richness-mediated effect).

H2: Dams fragment the lanscape but homogeneize biotic community

We distinguish a *local* pathway (distance to nearest dam) from a *global* pathway (cumulative barrier effect / basin-wide fragmentation). At the (global) basin scale, cumulative (linear) downstream fragmentation is expected to reduce species richness, mean trophic level and trophic complementarity, through loss of migratory and piscivorous species that depend on longitudinal connectivity (Dean et al. 2023; He et al. 2024). On the other hand, the upstream cumultative fragmentation homogeneize hydrological conditions resulting in more similar biotic community (Poff et al. 2007). As a result, cumulative (dendritic) upstream fragmentation is expected to reduce trophic complementarity and increase connectance through the promotion of generalist – sometimes invasive – species (Johnson et al. 2008; Encina et al. 2024; Bylak et al. 2023). At a local scale, this phenomenon is expected to increase near reservoir and so should be come stronger as the distance to the nearest downstream dam decreases. Conversely, we expect the opposite effect of the distance to the nearest upstream dam because of the negative effect of sediment starvation and hydropeaking on productivity (Ellis and Jones 2013).

H3: The effect of dams and nutrients on food web structure is partly mediated by species richness

Dams and nutrients can impact species richness (as described above), which in turn impacts food web structure. This is an indirect pathway through which stressors can reshape food web. In addition, we predict candidate direct-effect mechanisms to operate. In particular, the impact of invasive species (described in H2) operates both indirectly and directly on food web structure. Indirectly through the potential exclusion of native species, directly through its unique generalist diet. We note, as a limitation, that the mediated effect from richness to food web structure may also capture the impact of other confounding factors acting on species richness and food web structure independently of stressors (such as habitat complexity).

H4: Stressor impacts change along the river-estuary-sea continuum, and peak in estuaries

We predict that stressor impacts (nutrient and dams) on food web structure — i.e. an interaction between continuum position and stressor magnitude — is not uniform along the continuum. Specifically, because estuaries are naturally stressful environments (fluctuating salinity, turbidity, low baseline diversity), resident communities may already sit closer to their tolerance limits and hence be more sensitive to added anthropogenic pressure (Elliott and Quintino 2007).

H5: Nutrient impact on food web structure is magnified in lentic, dam-impounded reaches

We predict a true stressor-by-stressor interaction: the effect of nutrient concentration on food web structure (H1) is amplified in sites closer to downstream dam because of longer residence time. Impoundment slows water transit, giving phytoplankton more time to take up available nutrients and convert them into biomass rather than being flushed downstream (Chen et al. 2020). We therefore expect the nutrient effect on species richness, connectance, mean trophic level and trophic complementarity to strengthen as distance to the nearest downstream dam decreases, i.e. a nutrient × local-downstream-dam interaction. This tests for a synergestic interaction between nutrient and dam impacts (Jackson et al. 2021; Orr et al. 2024).

Statistical model

We formalise H1–H5 as two linked Bayesian spatial models fit in INLA, following the mediation structure introduced in H3. Structural metrics (connectance, mean trophic level, trophic complementarity) are fit as separate response models sharing the same covariate structure. Site-level covariates are: `nutrient` (nutrient concentration or composite index, H1); `dam_down_local` and `dam_down_global` (distance to nearest downstream dam, and cumulative downstream fragmentation, H2); `dam_up_local` and `dam_up_global` (distance to nearest upstream dam, and cumulative upstream fragmentation, H2); `richness` (local species richness, H3); and `habitat` (categorical variable distinguishing *river*, *estuary* and *sea* sites). `S(site)` denotes the spatial random effect.

> **Note**
>
> `S(site)` can be a Whittle–Matérn field on the river metric graph (Bolin et al. 2023) for river sites, and a standard 2D SPDE field for coastal sites (if needed, maybe a metric graph will be sufficient if sites are close enough to the river mouth).

> **Note**
>
> We use a categorical variable to caracterise site position along the continuum to avoid identifiability issue. As a continuous variable (e.g. distance to river mouth) may already be captured by the spatial random effect, making the model unable to disentangle the two.

```
structure ~ nutrient + nutrient^2 # H1
```

```
              + dam_down_local + dam_down_global # H2
              + dam_up_local + dam_up_global # H2
              + nutrient:dam_down_local # H5
              + habitat:nutrient # H4
              + habitat:dam # H4
              + richness # H3
              + S(site) # Handle spatial auto-correlation

# H3 - Model richness as a mediator variable
richness ~ nutrient + nutrient^2
             + dam_down_local + dam_down_global
             + dam_up_local + dam_up_global
             + nutrient:dam_down_local
             + habitat:nutrient
             + habitat:dam
             + S(site)
```

> **Note**
>
> The `structure` and `richness` equations above are not a single joint model. Instead, we fit the two models separately. For H3's mediated pathway, we draw matched posterior samples of the relevant stressor coefficient from the `richness` fit and the `richness` coefficient from the `structure` fit, and multiply them to obtain a posterior distribution for the indirect effect. The direct effect is read directly from the `structure` model's stressor coefficient, and the total effect is the sum of the two. This approach is common practice in Bayesian mediation, but it is important to note that it does not fully propagate joint uncertainty.

Methods

Food web inference

To infer local food web structure, we used a metaweb approach. That is, we build a web compiling the potential feeding interactions between all species of the dataset. Importantly, our metaweb takes into account intra-specific variation in species diet. We did so by defining trophic interactions at different life stages for each species.

Metaweb nodes

The nodes of our metaweb represent a species at a given lifestage. Specifically, fish species were divided into nine body class sizes evenly distributed. Each body size class within a given species is what we call a 'trophic species' assuming that conspecifics of the same body size class share similar trophic interactions because trophic interactions are largely determined by predator–prey body size ratio in freshwater ecosystems (Brose et al. 2006).

In addition of fish species, seven basal nodes (resources) were added at the base of the metaweb: detritus, biofilm, phytoplanckton, zooplanckton, macrophytes and zoobenthos. We represent the basal food web structure in the Appendix (see Figure S1). We assumed that these resources were present at all sampling sites.

Infer missing fish lengths

Most length of fishes are measured but not all. Yet, we need individual fish length to determine their diet based on their ontogeny. We infer missing fish size using information on the given fishing operation. When there are more than three observations of the fish species, we use a truncated normal distribution (package truncdist; Nadarajah and Kotz (2006)). The mean and variance are given by the modes of the observed distribution of the fishing operation. The minimum and maximum of the distribution are given by the minimal and maximal length observed on the entire data set for that species. We do so to avoid infering irrealistic lengths.

Get fish diet

We use fishbase as a primary source to extract fish diet at their different life stages, to get on which basal categories species feed on and whether they are piscivorous or not. We detail in the Supporting Information how made the correspondance between our categories and the one of fishbase. We remove species that have an occurence lower than 10 in the entire dataset which comprise 69 species. This enable to avoid retrieving diet information of species that will have a minimal impact on food web structure due to their rarity. We add a larvae stage when missing, assuming that fish larvae feed on zooplankton (Dabrowski 1984). We assume that fish individuals are larvae if they measure less than 2 cm. To classify individuals as juvenile or adult we use the length at maturity of the species as a threshold. When this information was not available we used a single stage (*combined*) by merging the diets of the two stages. We did not consider a recruit stage.

The metaweb

Predation window are assumed to be 0.03 to 0.45 of individual body size for piscivorous species. We then use the package foodwebbuilder to build the metaweb and the local food webs through subsampling. We investigated the effect of the number of size classes on the structure of the reconstructed metaweb. We observed little influence of the number of size classes on the metaweb connectance Figure S3. So, we decided to split each species 5 size classes as in Bonnaffé et al. (2024a).

> **Note**
>
> The choice of a constant predation window has an important implication. It implies that what only matters in terms of ecosystem functioning is the individual size (once the species is known to be piscivorous). In this setting, two piscivorous individual of different species but same size will eat the very same things.

Subsampling the metaweb to get local food webs

Results

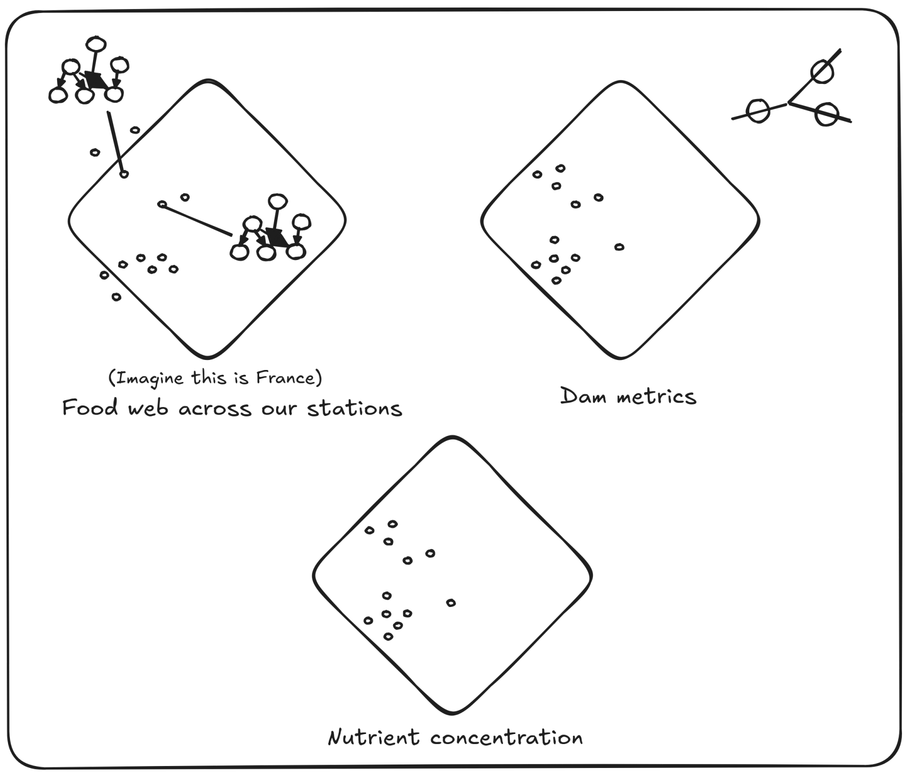

Figure 2. Map of our reconstructed food web and stressors.

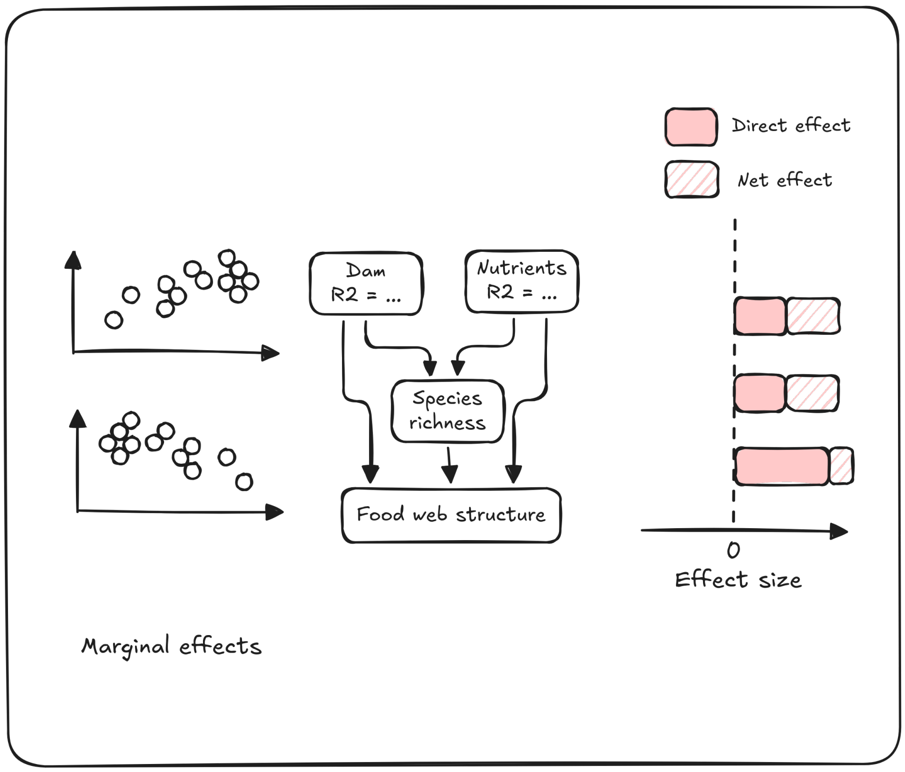

Fig 3. SEMs

Discussion

Limitations

Regarding food web structure, we have simplified the base of the web and exclude macro-invertebrates. This choice naturally emphasizes the functional role of fishes in shaping food web structures. Yet, it would be valuable to include macro-invertebrates within the food web to get a more exhaustive picture of how aquatic community respond to stressors along the riversea continuum.

Supporting Information

Structure of the food web base

Model DAG

Put model DAG here using targets.

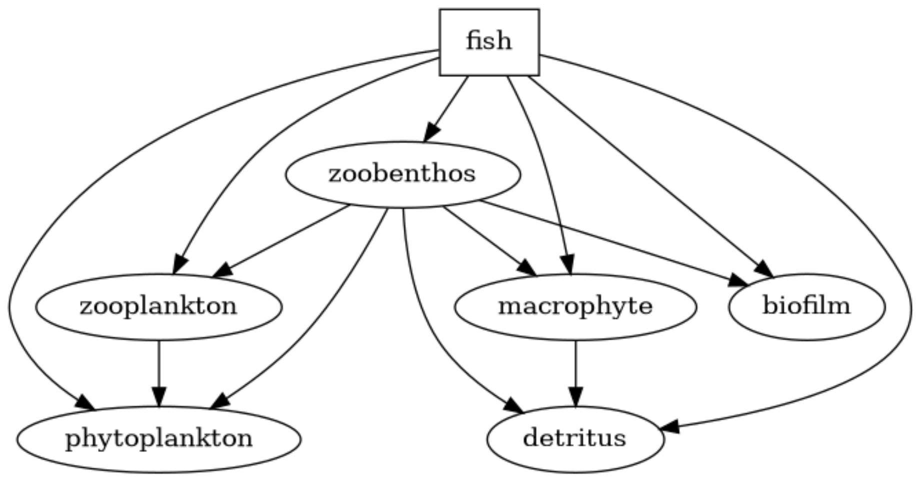

Figure S1: Assumed structure of the food web base.

Venn diagramm

Number of size classes and metaweb structure

Altermatt, Florian. 2013. "Diversity in Riverine Metacommunities: A Network Perspective." *Aquatic Ecology* 47 (3): 365–77. https://doi.org/10.1007/s10452-013-9450-3.

Altermatt, Florian, and Emanuel A. Fronhofer. 2018. "Dispersal in Dendritic Networks: Ecological Consequences on the Spatial Distribution of Population Densities." *Freshwater Biology* 63 (1): 22–32. https://doi.org/10.1111/fwb.12951.

Altermatt, Florian, Mathew Seymour, and Nicolas Martinez. 2013. "River Network Properties Shape -Diversity and Community Similarity Patterns of Aquatic Insect Communities Across Major Drainage Basins." *Journal of Biogeography* 40 (12): 2249–60. https://doi.org/10.1111/jbi.12178.

Álvarez-Romero, Jorge G., Robert L. Pressey, Natalie C. Ban, and Jon Brodie. 2015. "Advancing Land-Sea Conservation Planning: Integrating Modelling of Catchments, Land-Use Change, and River Plumes to Prioritise Catchment Management and Protection." *PLOS ONE* 10 (12): e0145574. https://doi.org/10.1371/journal.pone.0145574.

Attrill, Martin J. 2002. "A Testable Linear Model for Diversity Trends in Estuaries." *Journal of Animal Ecology* 71 (2): 262–69. https://doi.org/10.1046/j.1365-2656.2002.00593.x.

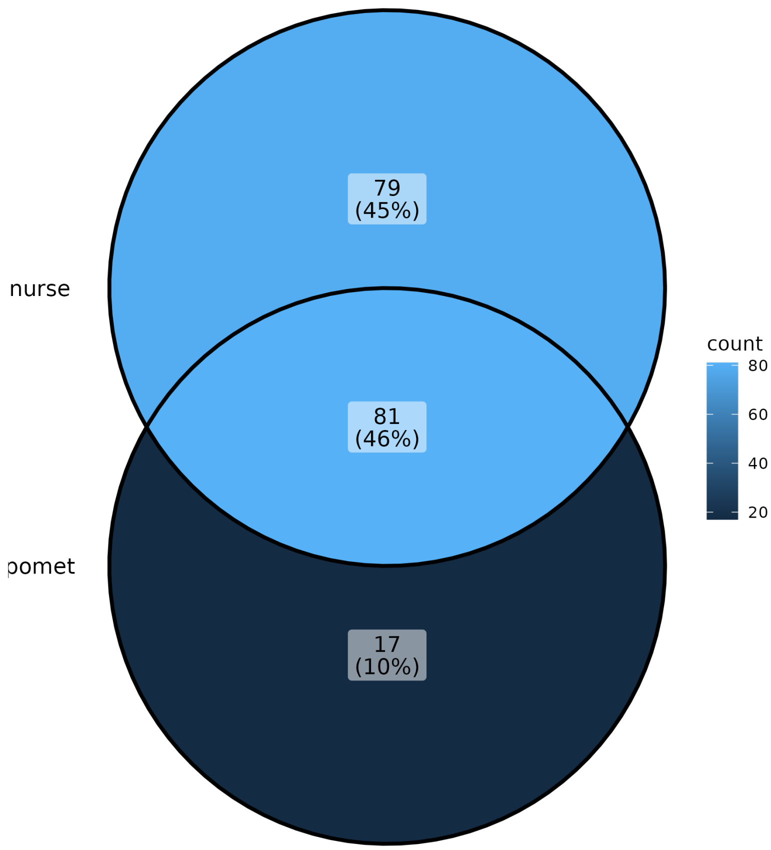

Figure S2: Venn diagram of common species between surveys.

Barnes, Andrew D., Ulrich Brose, Nico Eisenhauer, et al. 2026. "Food Web Complexity Underlies Biodiversity Effects on Ecosystem Functioning." *Nature*, July 1, 1–7. https://doi.org/10.1038/s41586-026-10710-5.

Bierschenk, Antje M., Candida Savage, and Christoph D. Matthaei. 2017. "Intensity of Catchment Land Use Influences Biological Traits of Benthic Invertebrates Along a Freshwater-Marine Continuum." *Limnology and Oceanography* 62 (S1): S292–308. https://doi.org/10.1002/lno.10584.

Binzer, Amrei, Christian Guill, Ulrich Brose, and Björn C. Rall. 2012. "The Dynamics of Food Chains Under Climate Change and Nutrient Enrichment." *Philosophical Transactions of the Royal Society B: Biological Sciences* 367 (1605): 2935–44. https://doi.org/10.1098/rstb.2012.0230.

```r
targets::tar_read(plot_metaweb_connectance)
```

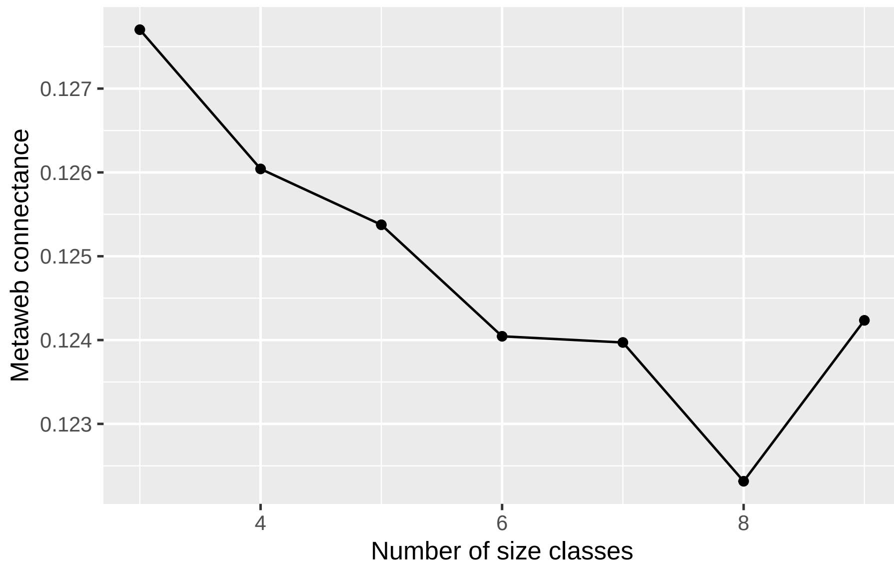

Figure S3: How metaweb connectance varies with the number of size classes.

Bolin, David, Alexandre B. Simas, and Jonas Wallin. 2023. *Gaussian Whittle-Matérn Fields on Metric Graphs*. https://doi.org/10.48550/arXiv.2205.06163.

Bonnaffé, Willem, Alain Danet, Camille Leclerc, Victor Frossard, Eric Edeline, and Arnaud Sentis. 2024b. "The Interaction Between Warming and Enrichment Accelerates Food-Web Simplification in Freshwater Systems." *Ecology Letters* 27 (8): e14480. https://doi.org/10.1111/ele.14480.

Bonnaffé, Willem, Alain Danet, Camille Leclerc, Victor Frossard, Eric Edeline, and Arnaud Sentis. 2024a. "The Interaction Between Warming and Enrichment Accelerates Food-Web Simplification in Freshwater Systems." *Ecology Letters* 27 (8): e14480. https://doi.org/10.1111/ele.14480.

Brose, Ulrich, Richard J. Williams, and Neo D. Martinez. 2006. "Allometric Scaling Enhances Stability in Complex Food Webs." *Ecology Letters* 9 (11): 1228–36. https://doi.org/10.1111/j.1461-0248.2006.00978.x.

Brosse, Sébastien, Nicolas Charpin, Guohuan Su, et al. 2021. "FISHMORPH: A Global Database on Morphological Traits of Freshwater Fishes." *Global Ecology and Biogeography* 30 (12): 2330–36. https://doi.org/10.1111/geb.13395.

Bylak, Aneta, Krzysztof Kukuła, Ryan Easton, Malcolm Reid, and Gerard P. Closs. 2023. "Reservoirs Facilitate Colonization of River Catchments by a Native Invasive Fish Through Provision of Pelagic Larval Rearing Habitat." *Biological Invasions* 25 (5): 1541–59. https://doi.org/10.1007/s10530-023-02995-8.

Carrara, Francesco, Florian Altermatt, Ignacio Rodriguez-Iturbe, and Andrea Rinaldo. 2012. "Dendritic Connectivity Controls Biodiversity Patterns in Experimental Metacommunities." *Proceedings of the National Academy of Sciences* 109 (15): 5761–66. https://doi.org/10.1073/pnas.1119651109.

Carraro, Luca, and Hsi-Cheng Ho. 2025. "Mechanistic Insights on Riverine Meta-Ecosystems: Network Shape Drives Spatial Biodiversity and Trophic Structures." *Ecohydrology* 18 (2): e2683. https://doi.org/10.1002/eco.2683.

Chalmandrier, Loïc, David Cunillera-Montcusí, Naiara López-Rojo, et al. 2025. "Natural Disturbances and Connectivity Drive Seasonal Taxonomic and Trait Patterns of Aquatic Macroinvertebrate Communities Across Europe." *Global Ecology and Biogeography* 34 (5): e70047. https://doi.org/10.1111/geb.70047.

Chan, Jeffery C. F., Billy Y. K. Lam, David Dudgeon, and Jia Huan Liew. 2025. "Global Consequences of Dam-Induced River Fragmentation on Diadromous Migrants: A Systematic Review and Meta-Analysis." *Biological Reviews* 100 (5): 2020–37. https://doi.org/10.1111/brv.70032.

Chen, Qiuwen, Wenqing Shi, Jef Huisman, et al. 2020. "Hydropower Reservoirs on the Upper Mekong River Modify Nutrient Bioavailability Downstream." *National Science Review* 7 (9): 1449–57. https://doi.org/10.1093/nsr/nwaa026.

Cohen, J E, and F Briand. 1984. "Trophic Links of Community Food Webs." *Proceedings of the National Academy of Sciences* 81 (13): 4105–9. https://doi.org/10.1073/pnas.81.13.4105.

Comte, Lise, and Julian D. Olden. 2018. "Fish Dispersal in Flowing Waters: A Synthesis of Movement- and Genetic-Based Studies." *Fish and Fisheries* 19 (6): 1063–77. https://doi.org/10.1111/faf.12312.

Dabrowski, K. 1984. "The Feeding of Fish Larvae : Present « State of the Art » and Perspectives." *Reproduction Nutrition Développement* 24 (6): 807–33. https://doi.org/10.1051/rnd:19840701.

Danet, Alain, Maud Mouchet, Willem Bonnaffé, Elisa Thébault, and Colin Fontaine. 2021. "Species Richness and Food-Web Structure Jointly Drive Community Biomass and Its Temporal Stability in Fish Communities." *Ecology Letters* 24 (11): 2364–77. https://doi.org/10.1111/ele.13857.

Dean, E. M., Dana M. Infante, Hao Yu, Arthur Cooper, Lizhu Wang, and Jared Ross. 2023. "Cumulative Effects of Dams on Migratory Fishes Across the Conterminous United States: Regional Patterns in Fish Responses to River Network Fragmentation." *River Research and Applications* 39 (9): 1736–48. https://doi.org/10.1002/rra.4173.

Death, Russell G., and Michael J. Winterbourn. 1995. "Diversity Patterns in Stream Benthic Invertebrate Communities: The Influence of Habitat Stability." *Ecology* 76 (5): 1446–60. https://doi.org/10.2307/1938147.

Ding, Na, Zunyi Xie, Xiaoming Wang, et al. 2026. "Freshwater Biodiversity Impacts of Global Hydropower Dams." *Communications Earth & Environment*, ahead of print, February 9. https://doi.org/10.1038/s43247-026-03263-y.

Dolan, Ellen J., Ismael Soto, Julian D. Olden, et al. 2026. "Native and Non-Native Populations Respond Unevenly to River Barrier Removals." *Global Change Biology* 32 (6): e70941. https://doi.org/10.1111/gcb.70941.

Doretto, Alberto, Elena Piano, and Courtney E. Larson. 2020. "The River Continuum Concept: Lessons from the Past and Perspectives for the Future." *Canadian Journal of Fisheries and Aquatic Sciences* 77 (11): 1853–64. https://doi.org/10.1139/cjfas-2020-0039.

Elliott, Michael, and Victor Quintino. 2007. "The Estuarine Quality Paradox, Environmental Homeostasis and the Difficulty of Detecting Anthropogenic Stress in Naturally Stressed Areas." *Marine Pollution Bulletin* 54 (6): 640–45. https://doi.org/10.1016/j.marpolbul.2007.02.003.

Ellis, Lucy Eunsun, and Nicholas Edward Jones. 2013. "Longitudinal Trends in Regulated Rivers: A Review and Synthesis Within the Context of the Serial Discontinuity Concept." *Environmental Reviews* 21 (3): 136–48. https://doi.org/10.1139/er-2012-0064.

Encina, Lourdes, Amadora Rodríguez-Ruiz, Carlos Orduna, Juan Ramón Cid, de Meo Ilaria, and Carlos Granado-Lorencio. 2024. "Impact of Invasive European Catfish (Silurus Glanis) on the Fish Community of Torrejón Reservoir (Central Spain) During a 11-Year Monitoring Study." *Biological Invasions* 26 (3): 745–56. https://doi.org/10.1007/s10530-023-03204-2.

Freitas, Leonardo Almeida, and Francisco Gerson Araújo. 2025. "Assessing Fish Community Structure and Diversity Across Environmental Gradients in a Tropical Bay." *Marine Ecology* 46 (2): e70016. https://doi.org/10.1111/maec.70016.

Frelat, Romain, Susanne Kortsch, Ingrid Kröncke, et al. 2022. "Food Web Structure and Community Composition: A Comparison Across Space and

Time in the North Sea." *Ecography* 2022 (2). https://doi.org/10.1111/ecog.05945.

Gounand, Isabelle, Nicolas Mouquet, Elsa Canard, Frédéric Guichard, Céline Hauzy, and Dominique Gravel. 2014. "The Paradox of Enrichment in Metaecosystems." *The American Naturalist* 184 (6): 752–63. https://doi.org/10.1086/678406.

Grill, G., B. Lehner, M. Thieme, et al. 2019. "Mapping the World's Free-Flowing Rivers." *Nature* 569 (7755): 215–21. https://doi.org/10.1038/s41586-019-1111-9.

Harris, Holly A. L., Tara J. Murray, Jonathan D. Tonkin, and Angus R. McIntosh. 2026. "Asynchronous River Floodplain Environment Dampens Ecological Variability Across Scales." *Oikos* 2026 (7). https://doi.org/10.1002/oik.11967.

He, Fengzhi, Christiane Zarfl, Klement Tockner, et al. 2024. "Hydropower Impacts on Riverine Biodiversity." *Nature Reviews Earth & Environment* 5 (11): 755–72. https://doi.org/10.1038/s43017-024-00596-0.

Ho, Hsi-Cheng, Florian Altermatt, and Luca Carraro. 2023. "Coupled Biological and Hydrological Processes Shape Spatial Food-Web Structures in Riverine Metacommunities." *Frontiers in Ecology and Evolution* 11 (June). https://doi.org/10.3389/fevo.2023.1147834.

Hong, Huasheng, Nengwang Chen, and Deli Wang. 2015. "River-Estuary-Coast Continuum: Biogeochemistry and Ecological Response to Increasing Human and Climatic Changes – Editorial Overview." *Estuarine, Coastal and Shelf Science*, River-estuary-coast continuum: Biogeochemistry and ecological response to increasing human and climatic changes, vol. 166 (December): 144–45. https://doi.org/10.1016/j.ecss.2015.10.036.

Humphries, Paul, Hubert Keckeis, and Brian Finlayson. 2014. "The River Wave Concept: Integrating River Ecosystem Models." *BioScience* 64 (10): 870–82. https://doi.org/10.1093/biosci/biu130.

Jackson, Michelle C., Samraat Pawar, and Guy Woodward. 2021. "The Temporal Dynamics of Multiple Stressor Effects: From Individuals to Ecosystems." *Trends in Ecology & Evolution* 36 (5): 402–10. https://doi.org/10.1016/j.tree.2021.01.005.

Johnson, Pieter TJ, Julian D Olden, and M Jake Vander Zanden. 2008. "Dam Invaders: Impoundments Facilitate Biological Invasions into Freshwaters." *Frontiers in Ecology and the Environment* 6 (7): 357–63. https://doi.org/10.1890/070156.

Junk, W. J., Peter Bayley, and Richard Sparks. 1989. "The Flood Pulse Concept in River-Floodplain Systems." *Canadian Special Publication of Fisheries and Aquatic Sciences* 106 (September): 110–27.

Kortsch, Susanne, Romain Frelat, Laurene Pecuchet, et al. 2021. "Disentangling Temporal Food Web Dynamics Facilitates Understanding of Ecosystem Functioning." *Journal of Animal Ecology* 90 (5): 1205–16. https://doi.org/10.1111/1365-2656.13447.

Lechêne, Alain, Philippe Boët, Pascal Laffaille, and Jérémy Lobry. 2018. "Nekton Communities of Tidally Restored Marshes: A Whole-Estuary Approach." *Estuarine, Coastal and Shelf Science* 207 (July): 368–82. https://doi.org/10.1016/j.ecss.2017.08.038.

LeCraw, Robin M., Pavel Kratina, and Diane S. Srivastava. 2014. "Food Web Complexity and Stability Across Habitat Connectivity Gradients." *Oecologia* 176 (4): 903–15. https://doi.org/10.1007/s00442-014-3083-7.

Leibold, M. A., M. Holyoak, N. Mouquet, et al. 2004. "The Metacommunity Concept: A Framework for Multi-Scale Community Ecology." *Ecology Letters* 7 (7): 601–13. https://doi.org/10.1111/j.1461-0248.2004.00608.x.

Little, S., P. J. Wood, and M. Elliott. 2017. "Quantifying Salinity-Induced Changes on Estuarine Benthic Fauna: The Potential Implications of Climate Change." *Estuarine, Coastal and Shelf Science*, ECSA 55 unbounded boundaries and shifting baselines: Estuaries and coastal seas in a rapidly changing world, vol. 198 (November): 610–25. https://doi.org/10.1016/j.ecss.2016.07.020.

Loreau, Michel, Nicolas Mouquet, and Robert D. Holt. 2003. "Meta-Ecosystems: A Theoretical Framework for a Spatial Ecosystem Ecology." *Ecology Letters* 6 (8): 673–79. https://doi.org/10.1046/j.1461-0248.2003.00483.x.

Ma, Fei, Hong Huang, Qi Yang, et al. 2026. "Biodiversity and Habitat Complexity Buffer the Destabilizing Effects of Anthropogenic Activities on Riverine Fish Communities." *Nature Communications*, ahead of print, May 18. https://doi.org/10.1038/s41467-026-73311-w.

Macedo, Morgana M., Ronaldo Angelini, VictorE. L. da Silva, and Nidia N. Fabré. 2021. "Trophic Structure of Coastal Meta-Ecosystems in the Tropical Southwestern Atlantic." *Estuarine, Coastal and Shelf Science* 263 (December): 107654. https://doi.org/10.1016/j.ecss.2021.107654.

Martinez, Neo D. 1992. "Constant Connectance in Community Food Webs." *The American Naturalist* 139 (6): 1208–18. https://www.jstor.org/stable/2462337.

McCann, K. S., J. B. Rasmussen, and J. Umbanhowar. 2005. "The Dynamics of Spatially Coupled Food Webs." *Ecology Letters* 8 (5): 513–23. https://doi.org/10.1111/j.1461-0248.2005.00742.x.

Mittelbach, Gary G., Christopher F. Steiner, Samuel M. Scheiner, et al. 2001. "What Is the Observed Relationship Between Species Richness and Productivity?" *Ecology* 82 (9): 2381–96. https://doi.org/10.1890/0012-9658(2001)082%5B2381:WITORB%5D2.0.CO;2.

Muneepeerakul, Rachata, Enrico Bertuzzo, Heather J. Lynch, William F. Fagan, Andrea Rinaldo, and Ignacio Rodriguez-Iturbe. 2008. "Neutral Metacommunity Models Predict Fish Diversity Patterns in Mississippi–Missouri Basin." *Nature* 453 (7192): 220–22. https://doi.org/10.1038/nature06813.

Mwanake, Ricky Mwangada, Elizabeth Gachibu Wangari, Daniel Graeber, and Ralf Kiese. 2026. "Diverging Riverine Dissolved c:n:p Ratios Under Global Environmental Change." *Npj Clean Water* 9 (1): 46. https://doi.org/10.1038/s41545-026-00596-1.

Nadarajah, Saralees, and Samuel Kotz. 2006. "R Programs for Truncated Distributions." *Journal of Statistical Software* 16 (August): 1–8. https://doi.org/10.18637/jss.v016.c02.

Nakano, Shigeru, and Masashi Murakami. 2001. "Reciprocal Subsidies: Dynamic Interdependence Between Terrestrial and Aquatic Food Webs." *Proceedings of the National Academy of Sciences* 98 (1): 166–70. https://doi.org/10.1073/pnas.98.1.166.

Olli, Kalle, Robert Ptacnik, Riina Klais, and Timo Tamminen. 2019. "Phytoplankton Species Richness Along Coastal and Estuarine Salinity Continua." *The American Naturalist* 194 (2): E41–51. https://doi.org/10.1086/703657.

Orr, James A., Samuel J. Macaulay, Adriana Mordente, et al. 2024. "Studying Interactions Among Anthropogenic Stressors in Freshwater Ecosystems: A Systematic Review of 2396 Multiple-Stressor Experiments." *Ecology Letters* 27 (6): e14463. https://doi.org/10.1111/ele.14463.

Ouellet, Valerie, Mathias J. Collins, John F. Kocik, et al. 2022. "The Diadromous Watersheds-Ocean Continuum: Managing Diadromous Fish as a Community for Ecosystem Resilience." *Frontiers in Ecology and Evolution* 10 (November). https://doi.org/10.3389/fevo.2022.1007599.

Pedersen, M. I., N. Jepsen, K. Aarestrup, A. Koed, S. Pedersen, and F. Økland. 2012. "Loss of European Silver Eel Passing a Hydropower Station." *Journal of Applied Ichthyology* 28 (2): 189–93. https://doi.org/10.1111/j.1439-0426.2011.01913.x.

Peller, Tianna, Isabelle Gounand, Marie-Josée Fortin, and Frédéric Guichard. 2026. "Spatially-Nested Topologies Stabilize Meta-Ecosystems via Cross-Scale Source-Sink Dynamics." *Ecology* 107 (4): e70375. https://doi.org/10.1002/ecy.70375.

Pillai, Pradeep, Andrew Gonzalez, and Michel Loreau. 2011. "Metacommunity Theory Explains the Emergence of Food Web Complexity." *Proceedings of the National Academy of Sciences* 108 (48): 19293–98. https://doi.org/10.1073/pnas.1106235108.

Poff, N. LeRoy, Julian D. Olden, David M. Merritt, and David M. Pepin. 2007. "Homogenization of Regional River Dynamics by Dams and Global Biodiversity Implications." *Proceedings of the National Academy of Sciences* 104 (14): 5732–37. https://doi.org/10.1073/pnas.0609812104.

Polis, Gary A., Wendy B. Anderson, and Robert D. Holt. 1997. "Toward an Integration of Landscape and Food Web Ecology: The Dynamics of Spatially Subsidized Food Webs." *Annual Review of Ecology, Evolution, and Systematics* 28 (Volume 28, 1997): 289–316. https://doi.org/10.1146/annurev.ecolsys.28.1.289.

Post, David M. 2002. "The Long and Short of Food-Chain Length." *Trends in Ecology & Evolution* 17 (6): 269–77. https://doi.org/10.1016/S0169-5347(02)02455-2.

Regnier, Pierre, Laure Resplandy, Raymond G. Najjar, and Philippe Ciais. 2022. "The Land-to-Ocean Loops of the Global Carbon Cycle." *Nature* 603 (7901): 401–10. https://doi.org/10.1038/s41586-021-04339-9.

Rosenzweig, Michael L. 1971. "Paradox of Enrichment: Destabilization of Exploitation Ecosystems in Ecological Time." *Science* 171 (3969): 385–87. https://doi.org/10.1126/science.171.3969.385.

Shepherd, Russell G., and Beverly N. Ellis. 1997. "Leonardo Da Vinci's Tree and the Law of Channel Widths – Combining Quantitative Geomorphology and Art in Education." *Journal of Geoscience Education* 45 (5): 425–27. https://doi.org/10.5408/1089-9995-45.5.425.

Spaak, Jurg W., Mirco Bundschuh, Martin H. Entling, Ralf B. Schäfer, and Ralf Schulz. 2026. "Is an Ecosystem Perspective Sufficient to Understand Meta-Ecosystem Processes? A Critical Reflection." *Oikos* 2026 (1): e11127. https://doi.org/10.1002/oik.11127.

Su, Guohuan, Adam Mertel, Sébastien Brosse, and Justin M. Calabrese. 2023. "Species Invasiveness and Community Invasibility of North American Freshwater Fish Fauna Revealed via Trait-Based Analysis." *Nature Communications*

14 (1): 2332. https://doi.org/10.1038/s41467-023-38107-2.

Tee, Hwee Sze, David Waite, Gavin Lear, and Kim Marie Handley. 2021. "Microbial River-to-Sea Continuum: Gradients in Benthic and Planktonic Diversity, Osmoregulation and Nutrient Cycling." *Microbiome* 9 (1): 190. https://doi.org/10.1186/s40168-021-01145-3.

Toussaint, Aurèle, Nicolas Charpin, Olivier Beauchard, et al. 2018. "Non-Native Species Led to Marked Shifts in Functional Diversity of the World Freshwater Fish Faunas." *Ecology Letters* 21 (11): 1649–59. https://doi.org/10.1111/ele.13141.

Truchy, Amélie, David Eme, Franck Jabot, et al. 2026. "Integrating Metaecosystem Framework with Ecosystem Service Dynamics in Sociohydrosystems." *BioScience* 76 (3): 238–53. https://doi.org/10.1093/biosci/biaf192.

Vannote, Robin L., G. Wayne Minshall, Kenneth W. Cummins, James R. Sedell, and Colbert E. Cushing. 1980. "The River Continuum Concept." *Canadian Journal of Fisheries and Aquatic Sciences* 37 (1): 130–37. https://doi.org/10.1139/f80-017.

Ward, J. V., and J. A. Stanford. 1995. "The Serial Discontinuity Concept: Extending the Model to Floodplain Rivers." *Regulated Rivers: Research & Management* 10 (2-4): 159–68. https://doi.org/10.1002/rrr.3450100211.

Whitfield, Alan K., Stephen J. M. Blaber, Michael Elliott, and Trevor D. Harrison. 2024. "Trophic Ecology of Fishes in Estuaries." *Reviews in Fish Biology and Fisheries* 34 (4): 1371–405. https://doi.org/10.1007/s11160-024-09878-8.

Xenopoulos, Marguerite A., John A. Downing, M. Dileep Kumar, Susanne Menden-Deuer, and Maren Voss. 2017. "Headwaters to Oceans: Ecological and Biogeochemical Contrasts Across the Aquatic Continuum." *Limnology and Oceanography* 62 (S1): S3–14. https://doi.org/10.1002/lno.10721.

Zhang, Song, Xiaowei Zhang, Wenjun Zhong, Lewkowicz Anna, Yan Zhang, and Loïc Pellissier. 2026. "Precipitation Modulates the Impact of Human Activities on Riverine Food Webs Revealed by Environmental DNA." *Ecology and Evolution* 16 (6): e73713. https://doi.org/10.1002/ece3.73713.

Zhu, Kai, Yiluan Song, Josephine C. Lesage, et al. 2024. "Rapid Shifts in Grassland Communities Driven by Climate Change." *Nature Ecology & Evolution*, October 16, 1–13. https://doi.org/10.1038/s41559-024-02552-z.